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F78AB
Actuarial and Financial Mathematics

Chapter 8: Yields

LEADERS IN **IDEAS** AND **SOLUTIONS**

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An Example

- Consider an investment that will pay you a cashflow of 1 at the end of each of the next four time units.
- What price should you be prepared to pay for this investment?
- If we were given an effective rate of interest, we could calculate the price of this investment using the Principle of Equivalence.

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An Example

- Suppose the effective rate of interest is 10% per time unit. Then the price, denoted P, is:

$$\boldsymbol{P = a_{\overline{4}|}^{0.1} = £3.17\ (2\ d.p.)}$$

- However, suppose that this investment is actually a financial contract that is traded in a financial market.
- The price of the investment will then be determined by the market at any given time.

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An Example

- Suppose that on some given date, the price is P = 3.000. Then a different question presents itself:
  - What rate of interest is consistent with a market price of 3.000 for this investment?
- This rate of interest, if it exists, is called the **yield** on the investment.

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Another Example

An investor pays £200 now in order to get £115 in 3 years' time and £115 in 6 years' time. What is the annual effective rate of interest earned on this investment?

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An Example

- For the earlier example we know that,

$$P = a_{\overline{4}|} = 3.170$$

  if the effective rate of interest per unit of time is 10%.
- What is the effective rate of interest per unit of time if *P* = 3.000? Should this yield be more or less than 10%?
- The yield can be found by solving:

$$a^{i}_{\overline{4}|} = 3.000 \quad \text{OR} \quad s^{i}_{\overline{4}|} = 3(1+i)^4$$

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Linear Interpolation

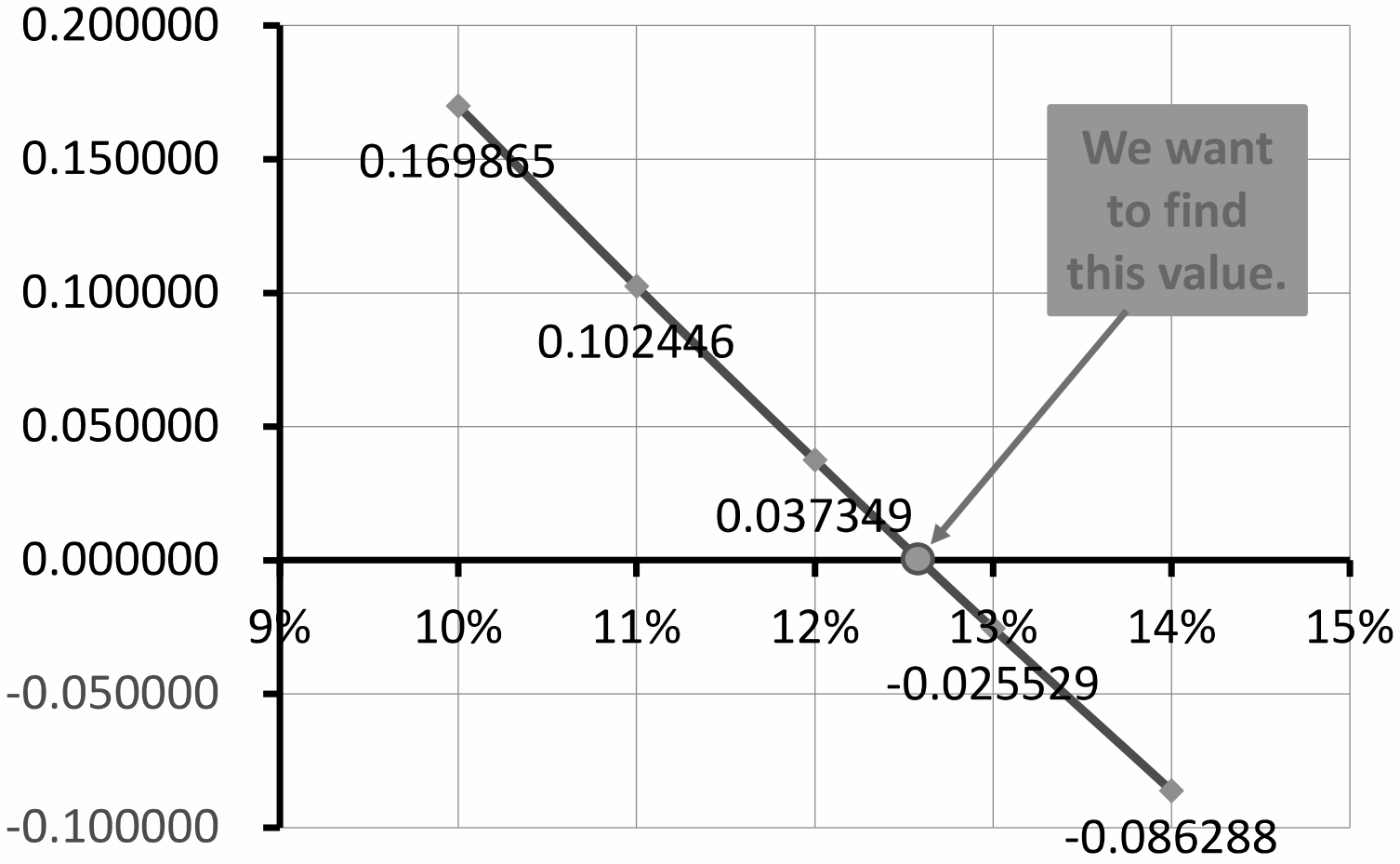
- We can use Linear Interpolation to solve

$$a\overline{4|}^{i} = 3.000 \quad \text{OR} \quad s\overline{4|}^{i} = 3\left(1+i\right)^{4}$$

- First we need to figure out the values between which i lies.
- Then we use these two values to get an approximate value for i.

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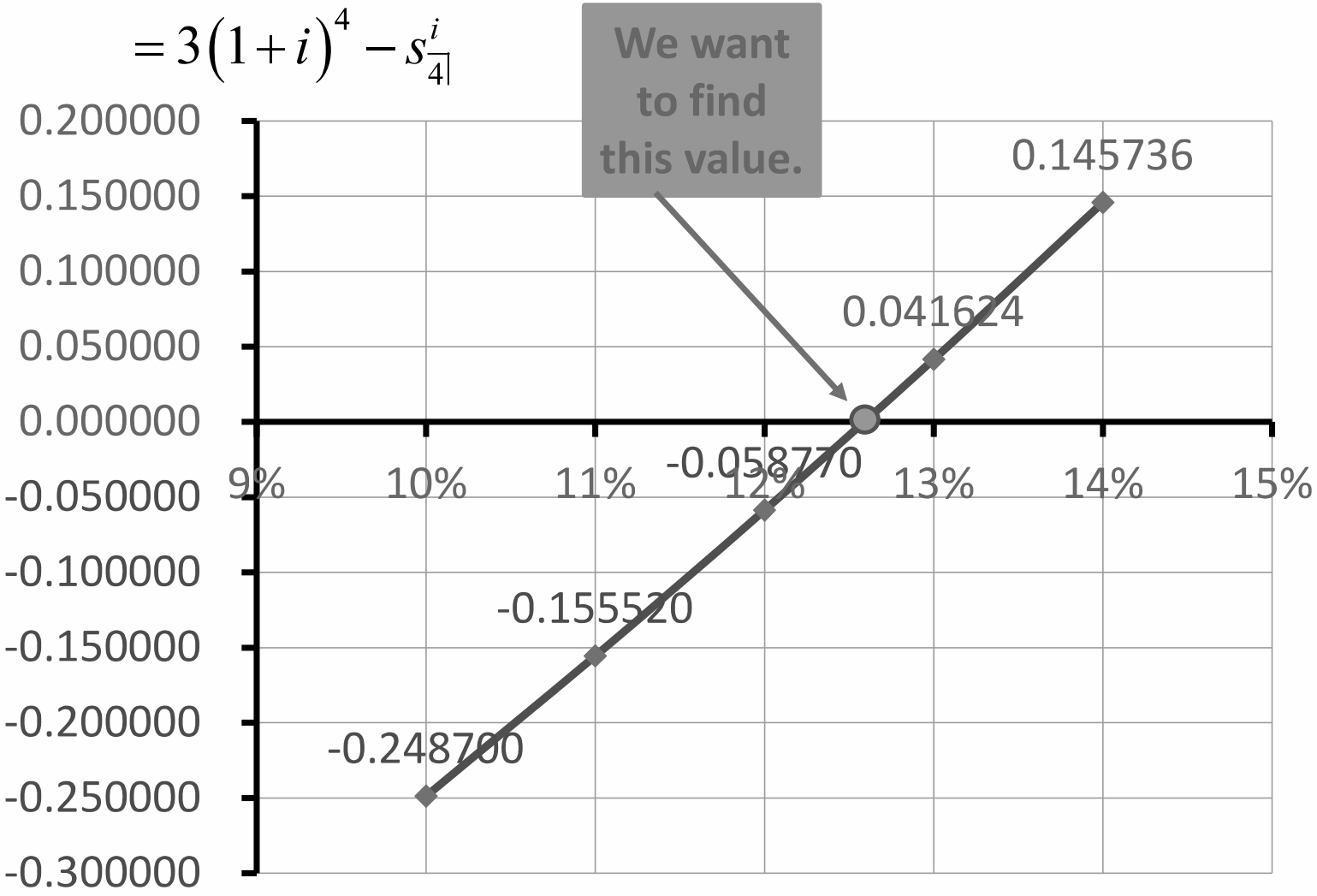
$$f(i) = a_{\overline{4}|}^{i} - 3 = v + v^2 + v^3 + v^4 - 3$$



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$$g(i) = 3(1+i)^4 - (1+i)^3 - (1+i)^2 - (1+i)^1 - (1+i)^0$$

$$= 3(1+i)^4 - s_{\overline{4}|}^{i}$$



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Solution to Example:

Let $g(i) = 3(1+i)^4 - s_{\overline{4}|}^{i}$. Then

- $g(0.12) = -0.058770$
- $g(0.13) = 0.041624.$

We then approximate the yield i^* by

$$i^* \approx 0.12 + 0.01 \times \frac{0.0588}{0.0588 + 0.0416} \approx 0.126$$

i.e. the yield is approximately 12.6%.

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Equations of Value

- To find the yield, construct an **equation of value** of the cashflows which is the equation involving a function of v or i.
- Say a financial contract has cashflows $c_1, c_2, c_3, \ldots c_n$ at times $t_1 < t_2 < t_3 \ldots < t_n$.
- We define

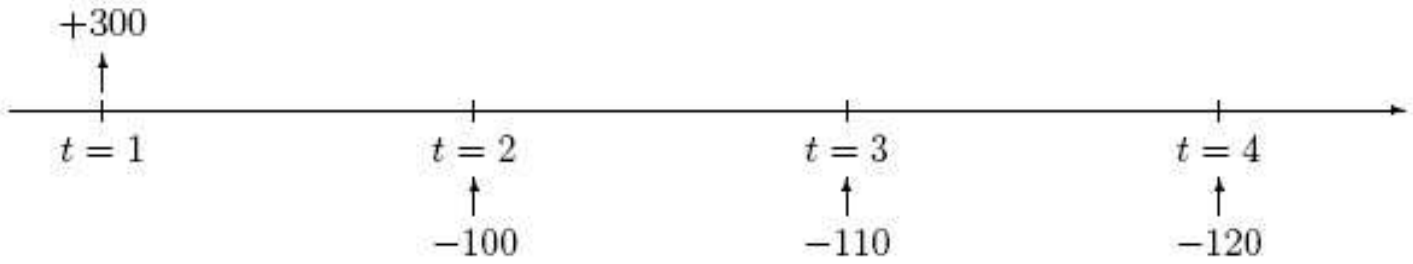
$$f(i) = \sum_{k=1}^{n} c_k (1+i)^{t_r - t_k}$$

which is the value of the cashflows at time t_r and f(i) = 0 is defined as the **equation of value**.

Equations of Value

E.g

What is the yield on the following set of cashflows?



De

$$f(i) = 300(1+i)^3 - 100(1+i)^2 - 110(1+i) - 120.$$

We need to solve $f(i) = 0$.

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Solution to Example:

First note that

$$300(1+3i) - 100(1+2i) - 110(1+i) - 120 = 0$$

has solution $i \approx 0.05$.

Using the $f(i)$ from the previous slide we have

- $f(0.04) = -5.1008$
- $f(0.05) = 1.5375.$

We then approximate the yield i^* by

$$i^* \approx 0.04 + 0.01 \times \frac{5.1008}{1.5375 + 5.1008} \approx 0.0477.$$

We can check that $f(0.0477) = -0.0042 \approx 0.$

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Equations of Value

You are offered an investment which will pay £1 at the end of each of the next 10 years, and £2 at the end of each of the following 10 years. The price of the investment is £15.

What is the yield?

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Existence and Uniqueness of Yields

Yields do not always exist

The equation of value,

$$f(i) = \sum_{r=1}^{n} c_r (1+i)^{t_n - t_r} = 0$$

which is a polynomial equation in i, can fail to provide a valid yield if:

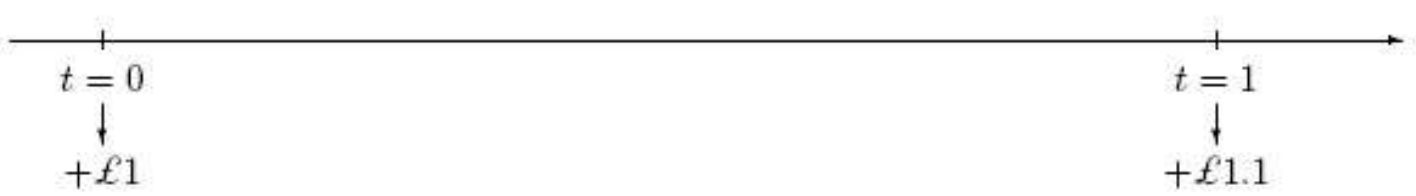
– There are no real solutions greater than or equal to -1. Either all the solutions are complex, or any real solutions are less than -1.

– There is more than one real solution greater than or equal to -1. Then we have no way of preferring one to the other as the yield.

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Existence and Uniqueness of Yields

Consider the following cashflows.



Here at time $t = 1$ we have

$$f(i) = (1 + i) + 1.1 = 0$$

i.e. $i = -2.1$, hence these cashflows do not have a yield.

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Existence and Uniqueness of Yields

Consider the following cashflows.



This gives

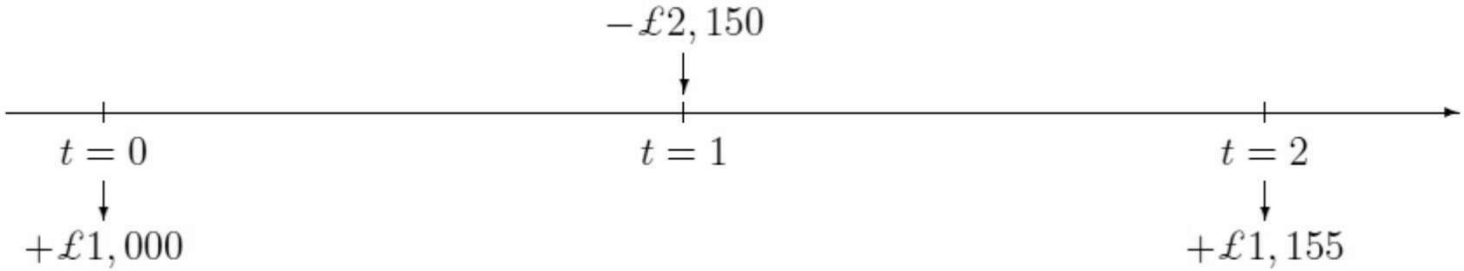
$$f(i) = (1+i)^2 + 1.1 = i^2 + 2i + 2.1 = 0$$

i.e. $i = -1 \pm \sqrt{-1.1}$, hence these cashflows do not have a yield.

Existence and Uniqueness of Yields

E.g.

Consider the following cashflows.



This gives

$$f(i) = 1000(1+i)^2 - 2150(1+i) + 1155 = 0$$

which factorises as $1000(i-0.1)(i-0.05) = 0$.
Hence $i = 0.1$ and $i = 0.05$ could be yields, hence the yield is undefined.

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Steffensen's Theorem

Say a financial contract has cashflows $c_1, c_2, c_3, \ldots c_n$ at times $t_1 < t_2 < t_3 \ldots < t_n$. If the cashflows change sign exactly once then the yield exists.

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Steffensen's Theorem

Sketch Proof

- Without loss of generality, suppose $c_1, c_2, c_3, \ldots c_k$ are negative cashflows and $c_{k+1}, c_{k+2}, c_{k+3}, \ldots c_n$ are positive cashflows.
- Define $$f(i) = \sum_{r=1}^{k} c_r(1+i)^{t_n - t_r} + \sum_{r=k+1}^{n} c_r(1+i)^{t_n - t_r}$$
- The equation $f(i) = 0$ is the equation of value.

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Steffensen's Theorem

Change to time t_k by multiplying through by $(1+i)^{t_k - t_n}$ and rewrite the equation of value as

$$\boxed{\sum_{r=1}^{k} c_r (1+i)^{t_k - t_r}} - \boxed{\sum_{r=k+1}^{n} -c_r (1+i)^{t_k - t_r}} = 0$$

Define the first sum as $g_1(i)$, and

the second as $g_2(i)$

so the equation of value is $g_1(i) - g_2(i) = 0$

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Steffensen's Theorem

- $g_1(i) < 0$, is a decreasing function of i,

$$\lim_{i \to -1} g_1(i) = c_k < 0 \quad \text{AND} \quad \lim_{i \to \infty} g_1(i) = -\infty$$

- $g_2(i) < 0$, is an increasing function of i,

$$\lim_{i \to -1} g_2(i) = -\infty \quad \text{AND} \quad \lim_{i \to \infty} g_2(i) = 0$$

- As $g_1(i)$ and $g_2(i)$ are continuous functions of i, the equation of value has exactly one solution in the interval [-1,∞).

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Prices and Yields

- Faced with several possible investment opportunities which will produce some future series of certain cashflows, how do we determine which investment we should choose to invest in?
- We need objective criteria to guide this choice.
- One possible criterion is to choose the investment opportunity with the largest yield.

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Prices and Yields

- There are two obvious questions that need to be answered:
  1. To earn a given yield, what price should we pay?
  2. If we pay a given price, what yield do we earn?

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Prices and Yields

Consider the two investments represented by the timelines given below. On the basis of yield, which do we prefer?

Investment A:



Investment B:



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Attendance matters!

